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Executive Summary

The core question is whether a deterministic **fractal** micro-dynamics (the IST "invariant set") can be rigorously coarse-grained into a smooth Bohmian fluid with velocity $$v=\frac{J}{\rho}=\frac{\hbar}{m}\Im! \frac{\nabla\psi}{\psi}\,. $$ In current mathematics, **no full proof exists**. However, several partial ingredients suggest it *may be possible in principle* under very strong assumptions: (1) By general results on continuity equations in metric spaces【48†L441-L449】【47†L583-L591】, any sufficiently regular limit of micro-trajectories yields a continuity equation $\partial_t\rho+\nabla\cdot J=0$ with some flux (or derivation) measure. (2) Under the *monokinetic ansatz* $J=\rho v$, one obtains a well-defined velocity field. (3) Stochastic-mechanical variational principles (Guerra–Morato) recover $v=\nabla S/m$ and the quantum potential【29†L101-L109】. (4) There are rigorous **hydrodynamic-limit** theorems (e.g. relative-entropy methods【36†L58-L64】) that derive fluid equations (Euler-type) from kinetic models under **monokinetic closure**.

Despite this, **critical gaps remain**. In particular, the existence of a smooth potential S (hence ψ) and single-valued phase is not automatic (Wallstrom's quantization issue【27†L25-L34】), and the fractal nature of I_U raises questions about defining continuous derivatives at all【31†L85-L93】. No existing theorem shows that an SRB-like measure on a fractal yield a *single-valued* velocity potential with $\nabla\times v=0$. Most relevant math (Dirichlet forms on fractals【31†L85-L93】, continuity-equation on metric spaces【48†L441-L449】, hydrodynamic limits for Vlasov/Euler【36†L58-L64】【47†L583-L591】) **falls short of proving** the BM-IST bridge. In sum, the idea is not *a priori* impossible, but it is **"POSSIBLY TRUE WITH MISSING PROOFS"**: it will require new theorems ensuring (i) convergence to a well-defined limiting (ρ,J), (ii) a monokinetic closure $J=\rho v$ on I_U, and (iii) emergence of a globally single-valued phase S.

Key points: Continuity-equation theory guarantees that any reasonable limit of ensemble trajectories yields $\partial_t\rho+\nabla\cdot J=0$【48†L441-L449】【47†L583-L591】. Hydrodynamic-limit results show that under strong closures one gets Euler or pressureless Euler equations【36†L58-L64】. Fractal-analysis (Dirichlet-form) techniques allow defining gradients/divergences on certain fractal attractors【31†L85-L93】. Stochastic variational principles can recover $v=\nabla S/m$ and thus the guidance law【29†L101-L109】. However, **no existing work simultaneously ensures all requirements for BM-IST**: in particular, *Wallstrom's objection* must be resolved, and one must prove that a continuous flow exists on the Cantor-like set I_U. Missing are rigorous proofs of an SRB-measure on I_U (the "Yang–Mills existence theorem" for the attractor), of the monokinetic hydrodynamic limit, and of Madelung quantization. Until such theorems are found, the BM-IST velocity derivation remains *unestablished*.

1. Mathematical Formulation

Target: A theorem of the form

> *Under suitable hypotheses on a fractal dynamical system $(I_U,\mathcal{T}_t,\mu)$, the coarse-grained density $\rho(t,x)$ and current $J(t,x)$ satisfy $\partial_t\rho+\nabla\cdot J=0$ with $J=\rho v$ for a unique velocity field $v(t,x)$. Moreover, in the regime analogous to non-relativistic QM,*

$v=(\hbar/m)\Im(\nabla\psi/\psi)$ for $\psi=\sqrt{\rho}\,e^{iS/\hbar}$ and S satisfying the Madelung (or Schrödinger) equations.

Here we must **define precisely** the objects and convergence:

- **Microscopic data**: Let $I_U\subset X$ be a compact fractal "state space" (e.g. a Cantor-type subset of a manifold) equipped with a continuous flow $\mathcal{T}_t:I_U\to I_U$ preserving a Borel probability measure μ (the SRB invariant measure). An initial point $u\in I_U$ represents a global universe state; under $\mathcal{T}_t(u)$ it traces a (deterministic) trajectory on I_U.

- **Embedding/observables**: There is a projection $x:I_U\to\mathbb{R}^3$ (the physical 3-space position of a particle subsystem) and a velocity function $v:I_U\to\mathbb{R}^3$ (given, for instance, by the tangent of the flow on I_U projected to space). Thus each u induces a physical trajectory $x(t)=x(\mathcal{T}_t(u))$.

- **Coarse-graining**: Consider an ensemble of micro-states u distributed by μ. Define the empirical mass density $$ \rho_\epsilon(t,x)\;=\;\int_{I_U}!K_\epsilon\big(x(\mathcal{T}*t(u)) - x\big)\, \mu(du), $$ where $K*\epsilon$ is a smooth mollifier of radius ϵ. Similarly define the microscopic current $$ J_\epsilon(t,x)\;=\;\int_{I_U}!v(\mathcal{T}*t(u))\,K*\epsilon\big(x(\mathcal{T}_t(u)) - x\big)\,\mu(du). $$ As $\epsilon\to0$, one obtains macroscopic fields (ρ,J) in the sense of distributions or measures.

- **Continuity equation**: Formally ρ_ϵ,J_ϵ satisfy $\partial_t\rho_\epsilon+ \nabla\cdot J_\epsilon=0$ for each ϵ, by construction of the particle flow. We aim to pass $ \epsilon\to0$ to get $\partial_t\rho+\nabla\cdot J=0$ in a weak sense.

- **Convergence modes**: One typically expects $\rho_\epsilon\to\rho$ narrowly (weak-*) in measures (or in $\mathcal{P}(\mathbb{R}^3)$) and $J_\epsilon\to J$ in the sense of vector-valued measures. A sufficiently strong convergence** (e.g. narrow plus uniform bounds) would ensure $J(t,\cdot)$ is absolutely continuous w.r.t. $\rho(t,\cdot)$ for a.e. t, so one can define the velocity density field $v(t,x)=J/\rho$. Precisely, a convenient framework is that $(\rho(t),J(t))$ solves the continuity equation in the sense of distributions, and $J=\rho v$ a.e. with $v\in L^1_{\rm loc}(\rho\,dt)$. We must then require that v is essentially bounded or L^p in x to ensure well-posedness of trajectories.

- **Hypotheses on $(I_U,\mathcal{T},\mu)$**: We seek minimal conditions (class $\mathcal{C}$) so that the above limit is guaranteed. Likely requirements include: *ergodicity/hyperbolicity* of the flow; *Ahlfors–regularity* of μ on I_U; the embedding $x(\cdot)$ is sufficiently smooth (say Lipschitz) on I_U; **monokinetic** dynamics so that multiple velocities don't accumulate at a point. The latter is a strong restriction analogous to requiring an underlying kinetic distribution collapse to a single velocity at each x. We also may need the coarse-grained μ_t to be an absolutely continuous (AC^1 or BV) curve in $\mathcal{P}(X)$.

- **Goal**: Show that in the limit (ρ,J) satisfies $J=\rho v$ and that $v(t,x)$ can be written $v=\nabla S/m$ for some scalar field $S(x,t)$ so that $\psi=\sqrt{\rho}\,e^{iS/\hbar}$ obeys Schrödinger's equation. Equivalently, derive the quantum Hamilton–Jacobi equation with quantum potential from the coarse dynamics.

Mathematically, the key questions are: 1. *Existence:* Does $(\rho_\epsilon,J_\epsilon)\to(\rho,J)$, and do they satisfy $\partial_t\rho+\nabla\cdot J=0$? 2. *Closure:* Is it true that $J(t,x)=\rho(t,x)\,v(t,x)$ for some v? (Equivalently, is the limiting flow *mono-kinetic*?) 3. *Uniqueness:* Under what convergence (e.g. narrow vs. L^p) do we get a unique v? 4. *Potential flow:* Can we show $\nabla\times v=0$, so $v=\nabla S/m$, and recover a single-valued phase (no circulation)? 5. *Quantum form:* If so, does (ρ,S) satisfy the Madelung (quantum) equations?

Precise theorem statements would thus take the shape:

> **Theorem (Conjectural).** *Let $(I_U,\mathcal{T}t,\mu)$ be a chaotic invariant set on which $x(\cdot)$ and $v(\cdot)$ are Lipschitz. Assume the ensemble density curve $t\mapsto\mu_t$ is W_1-absolutely continuous (or BV) in time. Then there exist limiting fields (ρ,J) solving $\partial_t\rho+\nabla\cdot J=0$. If furthermore the limit is "mono-kinetic" (no velocity dispersion), then $J=\rho v$ with $v\in L^1(\rho)$, then $v=\nabla S/m$ for some global phase S and the pair (ρ,S) satisfies the Madelung–Schrödinger equations.}(\rho)$, yielding a unique macroscopic velocity field. If additionally $\nabla\times v=0$ on $\mathrm{supp*

The weakest **mode of convergence** needed is that (μ_t) is **BV in the W_1-Wasserstein metric** on $\mathcal{P}(\mathbb{R}^3)$. By Theorem 1.4 of Abedi–Li–Schultz【48†L441-L449】【48†L461-L469】, this implies existence of a *flux measure* (ρ,J) or an equivalent *measure-valued derivation* solving the continuity equation. In practice, one would assume $\rho_\epsilon(t,x)\,dx\to\rho(t,x)\,dx$ narrowly and $J_\epsilon\to J$ weakly; ideally one has uniform moment bounds so that $J=\rho v$ as measures. If the convergence is stronger (e.g. $\rho_\epsilon\to\rho$ in $L^1_{\rm loc}$ and $J_\epsilon\to J$ in $L^1_{\rm loc}$), then $v=J/\rho$ is defined almost everywhere.

2. Relevant Literature

We review rigorous results bearing on this problem, organized thematically:

- **Continuity equation and metric-space limits:** Abedi–Li–Schultz (2026) give an abstract framework for continuity equations on metric-measure spaces【48†L441-L449】. **Theorem 1.4** (char. of BV-Wasserstein curves) shows that any W_1-BV curve of probability measures admits an equivalent description via either (B) a probability measure on path-space (the Lagrangian picture) or (C) a flux/derivation solution of the continuity equation【48†L441-L449】【48†L461-L469】. In $\mathbb{R}^n$, (C)↔(B) are fully equivalent【47†L583-L591】. In effect, *any suitable coarse-grained ensemble evolution has some flux J solving $\partial_t\rho+\nabla\cdot J=0$*. This guarantees existence of *some* velocity-like object, though not yet uniqueness or smoothness.

- **Hydrodynamic / kinetic limits:** There is a vast mathematical literature on deriving fluid equations from particle or kinetic models. Classical results (e.g. Dobrushin, Braun–Hepp) yield mean-field/Vlasov limits; more relevantly, relative-entropy methods have rigorously derived Euler or pressureless Euler systems under *monokinetic* closure assumptions【36†L58-L64】【20†L5380-L5388】. For example, Black–Tan (2024) prove that a kinetic "Cucker–Smale" flocking model under an *a priori* monokinetic ansatz converges to a compressible Euler-alignment system【36†L58-L64】. Han-Kwan–Michel (2021) rigorously derive Euler or Navier–Stokes limits of Vlasov–Stokes under heavy-friction monokinetic scalings【20†L5380-L5388】. These show that *if* one can force all velocities at a point to align (monokinetic), the limit obeys closed Euler equations. In our setting, one would hope that the chaos on I_U automatically enforces an effective monokinetic behavior. However, no general result guarantees that a deterministic fractal attractor yields $J=\rho v$ with a single v – indeed, most kinetic limits either *assume* monokinetic initial data or derive it from strong friction/dissipation. Some works on granular gases (Boltzmann eqn) indicate that generic solutions need not become monokinetic【34†L265-L274】.

- **Vector calculus on fractals:** A key concern is whether one can even define derivatives on a fractal attractor. There is a developing theory of "analysis on fractals" that partially addresses this. Hinz–Teplyaev (2026) show that for any *strongly local* Dirichlet form on a metric measure space (like many fractal spaces), one can define gradient and divergence operators【31†L85-L93】. In particular, on examples like the Sierpinski gasket, one has a measurable Riemannian structure and can talk about ∇f for f in the domain of the form【31†L85-L93】. This suggests that if μ is supported on a fractal with a regular Dirichlet form (e.g. a hyperbolic attractor), one may treat gradients in a weak sense. However, these "derivations" are typically only L^2-objects (not continuous fields), and do not straightforwardly yield classical vector fields on $\mathbb{R}^3$. Cheeger's theorem (1999) provides differentiable charts under strong assumptions (doubling, Poincaré), which are not satisfied by Cantor sets. Thus existing fractal calculus results imply *some* notion of $\nabla\cdot$ and ∇ on the attractor, but do **not** directly guarantee a smooth $v(x)$.

- **Stochastic variational approaches:** Guerra–Morato (1983) and related works showed that a **stochastic variational principle** can reproduce the Bohmian current and quantum potential. In particular, Guerra–Morato use stochastic control to show the current velocity is $\nabla S/m$ and the Hamilton–Jacobi equation acquires the quantum potential【29†L101-L109】. Nelson (1966) and Yasue (1981) similarly derive the Schrödinger equation from Brownian ensembles with a least-action principle. These results indicate a **bridge (c)**: if the chaotic motion on I_U can be effectively modelled as a stochastic process (with diffusion constant $\hbar$), then the hydrodynamic limit automatically yields $v=\nabla S/m$【29†L101-L109】. However, IST aspires to a deterministic (non-stochastic) microstate, so one must justify any passage to a stochastic effective description. In any case, these theories provide *consistency*: they imply that *if* one obtains any coarse-grained fluid, then requiring it to satisfy Nelson's or Yasue's variational principle yields Bohmian form.

- **Hyperuniformity and optimal point distributions:** Some arguments (outside strict mathematics) invoke "universal optimality" and hyperuniformity of the invariant set【40†L29-L38】【42†L22-L30】. Hyperuniform point processes have vanishing long-wavelength fluctuations【42†L22-L30】, a property found in special lattices and quasicrystals. Roca (2026) proved many self-similar tilings are hyperuniform【40†L29-L38】. If the I_U attractor were hyperuniform in 3-space, its coarse ρ would automatically be smooth. However, hyperuniformity is a *rigidity condition* not proven for chaotic attractors in general. It is an empirical test: one can compute the structure factor of a fractal sample and check if $S(0)=0$. Hyperuniformity would support monokinetic closure, but failing it would not strictly forbid $v=\nabla S$.

- **Madelung/Wallstrom issue:** Reddiger–Poirier (2022) analyze the Madelung PDEs and Wallstrom's objection【27†L25-L34】. They find that a consistent mathematical Madelung theory *may exist* if one can impose the quantization of circulation as a natural condition. In other words, even if (ρ,v) solves the Madelung equations, without an extra condition $\oint v\cdot dx=2\pi\hbar n$, one cannot derive a single-valued ψ. This is a known "quantization condition" gap: Madelung's equations by themselves are too weak. In the BM-IST context, one would need to show the invariant set dynamics enforce this condition (i.e. loops in I_U only allow quantized action increments). No existing theorem accomplishes this. Reddiger–Poirier conclude more mathematical work is needed to settle Wallstrom's critique【27†L25-L34】.

- **No-Go results:** The principal negative result is that **nothing in classical dynamics ensures monokinetic closure or irrotational flow** generically. Incompressible Euler fluids can develop vorticity even if starting potential. In deterministic particle systems (e.g. interacting gas), one typically gets a pressure tensor. Results on nonuniqueness of ODE flows for Sobolev vector fields 【49†L1-L9】 hint that irregular fields can fail uniqueness. Moreover, constructing a continuous vector field on a totally disconnected Cantor set is impossible in the classical sense, so defining v globally requires working in ambient $\mathbb{R}^n$ coordinates. Finally, any weak limit argument (e.g. narrow convergence) can lose information about J/ρ. One must show no oscillations or concentrations occur. At present there is **no rigorous disproof**, but the lack of known techniques makes the problem highly nontrivial.

Summary of sources: Key rigorous results include Abedi et al. (2026) for continuity equation on metric spaces【48†L441-L449】【47†L583-L591】, Guerra–Morato (1983) for stochastic variational derivation of Bohmian velocity【29†L101-L109】, Black–Tan (2024) and Han-Kwan–Michel (2021) for hydrodynamic Euler limits under monokinetic assumptions【36†L58-L64】【20†L5380-L5388】, Hinz–Teplyaev (2026) for calculus on fractals【31†L85-L93】, Torquato–Stillinger (2003) for hyperuniformity【42†L22-L30】, and Reddiger–Poirier (2022) on Madelung/Wallstrom issues【27†L25-L34】. We now analyze candidate bridges and obstacles in light of these.

3. Candidate Bridges

We classify potential routes by the documents' suggestions, assessing their feasibility:

- **(a) Deterministic coarse-graining / hydrodynamic limit:** This approach tries to apply rigorous mean-field/hydrodynamic limits to the deterministic fractal dynamics. The most relevant theorems here are:

- *Abedi–Li–Schultz (2026)*: shows that any W_1-BV curve (μ_t) in $\mathcal{P}(X)$ gives rise to a flux measure solving $\partial_t\mu+\nabla\cdot\nu=0$【48†L441-L449】. In $\mathbb{R}^3$, this means a measure-valued current ν exists. By Theorem 3.23【47†L583-L591】, flux and derivation formulations are equivalent. Thus, if (μ_t) from coarse-graining is BV-W_1, we immediately get *some* solution (ρ,J) to the continuity equation. **Strengths:** It covers any deterministic micro-trajectory ensemble under broad conditions. **Weaknesses:** It does not guarantee $J=\rho v$; in fact, it allows *singular* parts of the derivation. Minimal solutions【47†L573-L581】 remove divergence-free components, but that only handles "cancellation" in the flux, not multiple velocities. No general hydrodynamic theorem (e.g. DiPerna–Lions for Vlasov→Euler) exists for a fractal attractor. Any derivation of $J=\rho v$ would require a *pressureless* or monokinetic assumption, as in classical limits of Vlasov: e.g. if all microstates at a given x in physical space have the same v, then $J=\rho v$. Without a special argument (e.g. strong damping or forced alignment), this is a **major missing hypothesis**.

- **(b) Stochastic/hybrid effective description:** Here one supposes that the complicated deterministic chaos on I_U can be replaced by an effective stochastic process (Fokker–Planck) with diffusion $\hbar$. Then one can apply Nelson's stochastic mechanics, where each particle obeys an SDE whose ensemble yields the Schrödinger equation. The appeal is that Guerra–Morato【29†L101-L109】 and Yasue show such a formulation directly yields $v=\nabla S/m$. **Strengths:** It directly encodes Bohmian behavior via a variational principle. It effectively imports known quantum results. **Weaknesses:** It diverges from the IST premise of strict determinism: one must either assume some "hidden" noise or derive it from the fractal structure. If the fractal is conjectured to have a p-adic or ultrametric structure, one might argue it produces pseudostochastic behavior. But no theorems show a deterministic fractal flow gives rise to Brownian motion at large scales. In the math literature, one can sometimes show chaotic systems have diffusive limits (central limit theorems for Axiom-A flows), but these rely on hyperbolicity and mixing【11†L546-L554】. A proof that $(I_U,\mathcal{T})$ yields an SDE with correct diffusion constant is absent. In principle, this approach *could* bridge IST to Bohmian if one assumes the fractal chaos is "as if" stochastic (perhaps invoking Sinai–Ruelle–Bowen theory for Anosov flows). But it remains heuristic without a precise embedding of the deterministic state-space in a probabilistic model.

- **(c) Variational/Thermodynamic unification:** The idea here is that both the Madelung fluid and the SRB measure arise from extremal principles. Ruelle's thermodynamic formalism defines SRB measures as equilibrium states of a certain "action" (topological pressure). Guerra–Morato show Bohmian trajectories are extrema of a stochastic action【29†L101-L109】. If one could identify a single variational principle of which $\mu_{\rm frac}$ and the Bohmian action are two faces, then $v=\nabla S$ would follow by that principle. For example, one might try to show the coarse-graining of Ruelle's SRB measure yields a maximizer of a functional identical to Guerra's. **Strengths:** This stays within the deterministic framework and might derive ψ naturally as a generating function. **Weaknesses:** This is highly speculative: it requires formulating Ruelle's pressure on a fractal and Guerra's functional in the same space. The mathematics of connecting hyperbolic SRB measures to continuum PDEs is undeveloped. To our knowledge no rigorous result links an SRB measure to a hydrodynamic variational principle. This remains an open conceptual bridge rather than an established route.

- **(d) Fractal differential calculus:** One could attempt to do calculus *on the fractal I_U itself*, then project to $\mathbb{R}^3$. Works by Kigami, Strichartz, Teplyaev, etc., have developed Laplacians and vector calculus on specific fractals (Sierpinski, etc.). Hinz–Teplyaev【31†L85-L93】 summarize that on a metric measure space with a regular Dirichlet form, one can define a "gradient" operator as a derivation. In principle, if (I_U,μ) admits a Dirichlet form (perhaps the one induced by its chaotic flow), one can define local currents and divergence on I_U. Then one would need to show that these abstract fields coincide with coarse-grained ρ,J. **Strengths:** This stays entirely within the mathematical theory of fractals. **Weaknesses:** The existing fractal analysis results do not ensure smoothness or even continuity of v on the ambient space. They typically produce only $L^2(\mu)$ gradients. Moreover, translating a derivation on the fractal to a classical vector field $v(x)$ is nontrivial. Absent a concrete Riemannian structure on I_U, this approach lacks a direct link to $\psi(x)$ in $\mathbb{R}^3$. It also does not directly enforce the Madelung structure (one must still impose quantization).

Summary of candidate bridges: (a) can establish a continuity equation but needs a monokinetic assumption; (b) directly yields Bohmian form but assumes a Brownian limit; (c) is an intriguing variational idea with no precise theorem; (d) gives analytic tools on fractals but not the full fluid picture. The most immediately promising path is (a)+(c): use continuity-equation results【48†L441-L449】【47†L583-L591】 to get (ρ,J), then impose minimality/monokinetic closure (perhaps justified by ergodicity or extremality) to derive an Euler–Madelung PDE.

4. Disconfirming Results and Obstacles

We now mention results or arguments that challenge the prospect:

- **Wallstrom's objection (quantization):** As noted, purely hydrodynamic/Madelung formulations can permit non-integer circulation of the phase, which quantum mechanics disallows【27†L25-L34】. In other words, even if one obtains $v=\nabla S/m$ a.e., one must still impose $\oint v\cdot d\ell = (2\pi\hbar) \mathbb{Z}$. Reddiger–Poirier conclude that a satisfactory Madelung theory will need an extra quantization axiom【27†L25-L34】. In BM-IST, such quantization must somehow emerge from the fractal's topology (e.g. multi-valued phases on disconnected Cantor pieces?). Currently there is *no derivation* of this, so we count it as an **open obstacle**.

- **Vorticity/non-monokinetic flows:** General fluid limit theorems show that vorticity can survive or even amplify. For example, even if each micro-trajectory is continuous, the coarse-grained limit might have multiple velocity contributions at a point, analogous to having a non-zero pressure or vorticity. There are results (e.g. Alberti–Crippa [49†L1-L9]) that continuous vector fields can have non-unique trajectories on sets of full measure, hinting at irregular limiting behavior. In our context, if (ρ,J) is not monokinetic, then the velocity field is not uniquely defined. Some see this as a fatal flaw: if chaotic mixing on I_U produces a velocity *distribution* at each x, then v is not single-valued. We have no theorem specifically ruling this out for fractals, but it is a real risk.

- **Regularity on Cantor sets:** It is classically impossible to have a *smooth* curve on a Cantor set, because it has empty interior. One must always talk about measure-theoretic or weak fields. The only hope is defining $v(x)$ on the full $\mathbb{R}^3$ by extension. But if I_U is "nowhere dense" in state space, continuity in I_U does not imply continuity in ambient space. This suggests any emergent $v(x)$ might be very irregular (perhaps only Hölder). In practice, Bohmian velocity fields ∇S are usually well-behaved (if ψ is smooth). Ensuring that the fractal-induced v has sufficient smoothness is nontrivial.

- **No direct impossibility theorem:** We are not aware of a rigorous "no-go" theorem that strictly forbids deriving Bohmian velocity from fractal SRB measures. However, the lack of existing techniques (hydrodynamic limit from fractals, fractal-to-Madelung mapping) means the burden is on demonstrating consistency. In absence of a theorem, we cannot categorically rule it out, but the obstacles above are serious.

5. Missing Theorems

If current mathematics is insufficient, we identify the minimal new results needed:

- **SRB existence for I_U:** First, one needs a theorem guaranteeing that the hypothesized fractal I_U with measure μ actually exists under the IST axioms (hyperbolicity, factorization, regularity). This is analogous to the “Yang–Mills existence” problem: prove a chaotic attractor exists in the full p-adic setting with these properties. Without this, all else is moot.

- **Hydrodynamic limit theorem on fractals:** A rigorous theorem showing that for any $(I_U, \mathcal{T}*t,\mu)$ in class $\mathcal{C}$, the coarse-grained empirical measures converge (as $N\to\infty$ or $\epsilon\to0$) to a solution (ρ,J) of $\partial_t\rho+\nabla\cdot J=0$. Specifically, one needs to assume something like: (μ_t) is $BV*$. This is partially covered by Theorem 1.4【48†L441-L449】, but that theorem assumes }$, and prove $(\rho_\epsilon,J_\epsilon)\to(\rho,J)$ in $ \mathcal{D*a priori* the limit measure curve is BV. We need a theorem that *constructs* μ_t and shows it is BV-W_1 from the fractal dynamics.

- **Monokinetic closure theorem:** Given the above limit (ρ,J), prove that $J(t,x)=\rho(t,x)\,v(t,x)$ for some $v(t,x)$. Equivalently, show no “velocity dispersion” remains. In kinetic theory, this is often done by entropy methods or by assuming collisions drive distribution to a delta. In the deterministic fractal context, one could try to prove an analogous “dispersion decays” property. Formally, one needs a theorem: *If (μ_t) is a minimizer of some action (or a unique SRB), then it has the property that all mass flows coherently.* This is entirely open. Perhaps one could prove: if μ_t is an **extremal** (SRB) measure, then $\exists v$ such that $J=\rho v$ and v is unique.

- **Potential flow and quantization:** Even if $v=J/\rho$ exists, one must show it is irrotational: $ \nabla_x\times v=0$. This is automatic in one dimension or for scalar potentials, but in 3D requires an argument. A new theorem needed here would say: *under our axioms, the induced velocity has zero curl on the support of ρ.* This probably ties back to the assumed global variational principle: if $ (\rho,J)$ arise from extremizing Guerra’s action, then $\nabla\times v=0$ follows. Finally, a **Wallstrom theorem** is needed: it would say that the phase S obtained from integrating v is single-valued up to $2\pi\hbar\mathbb{Z}$. I.e. *the circulations $\oint v\cdot d\ell$ are quantized*. This is a purely topological statement about (I_U,μ) that must be added.

- **Strict convergence mode:** We should formalize what convergence ensures a unique v. Likely an L^p_{loc} convergence of J_ϵ and ρ_ϵ with $\rho>0$ a.e. (Then $v_\epsilon=J_\epsilon/\rho_\epsilon$ would converge in L^p as well.) A theorem could state: *If $ \rho_\epsilon\to\rho$ strongly in L^1_{loc} and $J_\epsilon\to J$ in L^1_{loc}, then $v_\epsilon=J_\epsilon/\rho_\epsilon$ has a subsequence converging to J/ρ a.e.* This is straightforward analytically but worth spelling out.

In summary, the **decisive missing theorem** is roughly:

> *Let $(\mu_t)_{t\in[0,T]}$ be the marginal measures of a deterministic fractal dynamics in $ \mathcal{P}(\mathbb{R}^3)$. Assume (μ_t) is bounded-variation in W_1. Then there exist $ \rho,J$ such that $\mu_t(dx)=\rho(t,x)\,dx$ and (ρ,J) solve $\partial_t\rho+\nabla\cdot J=0$. Moreover, if the measure μ on the attractor is SRB and (ρ,J) minimize an appropriate action, then $J=\rho\nabla S/m$ with S single-valued.*

A plausible **proof strategy** (sketch) would be: (1) Use Theorem 1.4【48†L441-L449】 and its converse to get (ρ,J) from a path measure. (2) Show uniqueness/minimality of that solution (Theorem 3.25【47†L593-L601】) enforces J/ρ well-defined. (3) Invoke a variational argument (à la Guerra–Morato) to show $ \nabla\times v=0$ and recover S. Each step faces hard technicalities: step (1) is partly done, step (2) needs a new “no pressure/no vorticity” lemma, step (3) requires a link to quantum action. If any step fails, new mathematics is needed.

6. Evidence Ledger

Claim	Theorem/ Result	Source(s)	Key Assumptions	Applicability to BM-IST	Status
BV W_1 ⇒ Continuity EQ solution: Any W_1-BV curve of measures yields a flux/ derivation solving $\partial_t\rho+ \nabla\cdot J=0$.	Theorem 1.4 (Abedi–Li–Schultz 2026) 【48†L441-L449】 【48†L461-L469】	Metric-space continuity eq	$(\mu_t)\in BV_{W_1}$	Coarse-grained μ_t assumed BV-W_1	PAID (by 【48】)
Flux⇔Derivation equivalence: On $\mathbb{R}^n$, any flux measure solution induces a unique measure-valued derivation, and vice versa.	Theorem 3.23 (Abedi–Li–Schultz 2026) 【47†L583-L591】	Continuity eq equivalence	Finite-mass flux/ derivation	Link between Eulerian/ Lagrangian pictures	PAID
Minimal solutions coincide: Minimal flux solutions correspond to minimal derivations.	Theorem 3.25 (Abedi–Li–Schultz 2026) 【47†L593-L601】	Continuity eq theory	As above (ℝ^n setting)	Could enforce $J=\rho v$ if minimal	PAID (but minimality w/ closure?)
Stochastic variational yields $v=\nabla S/m$: Nelson/ Guerra show current velocity is gradient of phase; quantum potential arises.	Guerra–Morato (PRD1983) 【29†L101-L109】	Stochastic Mechanics (SM)	Under SM assumptions (Brownian diff.)	Suggests form of v if process is Brownian	PAID

Claim	Theorem/ Result	Source(s)	Key Assumptions	Applicability to BM-IST	Status
Hydrodynamic limit of kinetic model: With monokinetic initial data, Vlasov→Euler (compressible or alignment Euler) holds.	Black–Tan (2024) 【36†L58-L64】 (Euler-alignment limit)	Euler from Vlasov (relative entropy)	Monokinetic ansatz; smooth interaction kernels	Demonstrates fluid limit under closure	PAID
SRB measures via extremal principle: SRB measures are equilibrium states (Ruelle formalism).	Ruelle's thermodynamic formalism (1970s)	Dynamical systems (textbook)	Axiom-A hyperbolicity, transitivity	Micro-measure $ \mu$ as equilibrium	PAID (theory background)
Gradient/ divergence on fractals: On a metric space with a strong Dirichlet form, one can define gradients and divergences.	Hinz–Teplyaev (arXiv 2026) 【31†L85-L93】	Fractal analysis	Strongly local Dirichlet form on (X, μ)	Allows differential calculus on I_U	PAID
Hyperuniformity definition: A point process is hyperuniform if number-variance grows like surface area of window【42†L22-L30】.	Torquato–Stillinger (PRE 2003)【42†L22-L30】	Statistical mechanics (RMP)	Large-window variance test	Criterion for ρ fluctuations being low	PAID
Hyperuniform substitution tilings: Many planar self-similar tilings (e.g. Penrose) are hyperuniform.	Roca (arXiv 2026)【40†L29-L38】	Dynamical systems (math.DS)	Substitution rules, aperiodic order	Motivates checking I_U for uniformity	PAID (analogy)

Claim	Theorem/ Result	Source(s)	Key Assumptions	Applicability to BM-IST	Status
Madelung/PDE well-posedness: A satisfactory well-posed theory of Madelung equations *may be* constructed to recover QM without ad hoc conditions.	Reddiger–Poirier (2022) 【27†L25-L34】	Mathematical physics	Standard Madelung PDE assumptions	Suggests Wallstrom's condition not fatal	PARTIAL (conjectural)
No general monokinetic theorem: (Implied) Without extra assumptions, a limit of many trajectories need not be mono-kinetic.	(Implied by kinetic theory)	Kinetic theory texts	Generic initial data	Monokinetic needed for $J=\rho v$	OWED (needs proof or assumption)
Wallstrom no-go: Madelung eqs permit non-quantized circulation; Schrödinger single-valuedness is extra.	Wallstrom's critique (1989,1994) (and [27] discussion)	QM foundations	Madelung PDE only	Quantization not automatic from fluid eq.	OWED (must impose condition)
Existence of fractal attractor: (Conjectured) The IST axioms imply a genuine hyperbolic attractor I_U exists.	– (Open; cited as IST postulate)	IST literature (Palmer et al.)	IST axioms on cosmic dynamics	Required before any coarse-graining	OWED (Yang–Mills problem)

Claim	Theorem/ Result	Source(s)	Key Assumptions	Applicability to BM-IST	Status
Empirical convergence: Needs minimal criteria (tightness, regularity) to pass $N\to\infty$ to PDE.	– (Standard fluid limit theory)	PDE/ Probability (Lions–Perthame)	Uniform moment/ entropy bounds	Standard but not proven for fractals	OWED (requires derivation)

Legend: **PAID** = theorem/result exists; **OWED** = hypothesized, not yet proved; **PARTIAL** = partial progress or conjectural.

7. Computational/Toy Models

Given the difficulty of a general proof, **numerical experiments** on simplified toy models can be informative:

- **Toy fractal attractor:** Construct a low-dimensional chaotic map or flow on e.g. the unit square whose attractor is a Cantor-like fractal (e.g. a 3D Lorenz attractor projected to 2D). Ensure it has an invariant measure μ. For example, use a known chaotic map (baker's map, coupled logistic maps) and then restrict to a Cantor set by selection.

- **Particle ensemble simulation:** Sample N points according to μ on the attractor and numerically evolve them under $\mathcal{T}_t$. At each time compute the empirical density and current on a spatial grid (e.g. by binning or kernel smoothing). Monitor convergence as N grows: does $\rho_N(t,x)$ approach a smooth function, and does J_N/ρ_N approach a stable field?

- **Structure factor $S(k)$:** Treat the sample points as a point process in space at a fixed time. Compute the structure factor $S(k)=\lim_{R\to\infty}\frac{1}{N}\Big|\sum_{i=1}^N e^{ik\cdot x_i}\Big|^2$ (or via FFT on the density). Check whether $S(k)\to0$ as $|k|\to0$ (hyperuniform) or tends to a constant. A vanishing $S(0)$ would suggest suppressed long-range fluctuations (as in crystals or special quasicrystals【42†L22-L30】).

- **Velocity field diagnostics:** From the empirical (ρ_N,J_N), compute $v_N=J_N/\rho_N$ on grid. Check whether $\nabla\times v_N\approx0$ (curl nearly zero) or small but nonzero. Also check $\oint v_N\cdot d\ell$ around loops: do they cluster near integer multiples of $2\pi\hbar$? These are discrete diagnostics of the Wallstrom condition.

- **Numerical trajectory ensemble:** Solve a simple one-dimensional fractal flow (e.g. tent map with an invariant Cantor distribution) and track a large ensemble of points on it to estimate a current. See if it matches the gradient of any scalar field.

- **Algorithmic ideas:** Use standard PDE/measure tools: e.g. compute Wasserstein distances between $\rho_N(t)$ at successive times to test BV-W_1 continuity. Compute Ruelle–Perron–Frobenius densities if possible to approximate SRB.

These experiments cannot prove anything, but can provide evidence for or against key features (monokineticity, hyperuniformity, irrotationality). They also help calibrate assumptions needed for rigorous theorems.

8. Final Verdict and Next Steps

Verdict: POSSIBLY TRUE, BUT UNPROVEN. In principle, a coarse-graining of a universally optimal fractal could produce the Bohmian velocity field (the guiding law)【29†L101-L109】【47†L583-L591】, but **critical steps lack proof**. The strongest existing route combines: (i) metric-continuity results which give $\partial_t\rho+\nabla\cdot J=0$【48†L441-L449】【47†L583-L591】; (ii) a monokinetic ansatz $J=\rho v$ motivated by dynamics akin to pressureless Euler; (iii) Guerra–Morato–Nelson variational methods to show $v=\nabla S/m$ and produce a Schrödinger structure【29†L101-L109】. Together, these imply the desired formula $v=(\hbar/m)\Im(\nabla\psi/\psi)$, but only **if one assumes** away the main gaps.

Decisive missing theorem: A precise mathematical statement of the required limit is:

> **Conjectural Theorem:** *Let (μ_t) be the pushforward of the IST fractal measure μ under the embedding into physical space. Assume (μ_t) is BV-W_1. Then as $\epsilon\to0$, the smoothed densities ρ_ϵ and currents J_ϵ converge in the sense of measures to a solution (ρ,J) of $\partial_t\rho+\nabla\cdot J=0$. If moreover the invariant measure μ is SRB and the solution is minimal, then $J=\rho\nabla S/m$ for a globally single-valued $S(x,t)$.*

Key missing assumptions for this theorem to hold in practice are: **(a)** existence of the attractor measure μ with required regularity (Ahlfors condition, mixing); **(b)** uniform bounds to ensure tightness and limit existence; **(c)** a proof that the solution is monokinetic/minimal so $v=J/\rho$ is a proper velocity. Additional assumptions needed include enough smoothness of $x(\cdot)$ and $v(\cdot)$ on I_U to make sense of ∇_x and ∇_v.

Fatal obstacles: Without proving that μ_t is BV in W_1, the continuity equation approach fails; without monokinetic closure, no single v exists. The Wallstrom quantization condition is not derivable from current methods. If any of these fail, the BM-IST program collapses back to an interpretation without derivation.

Next steps: Mathematicians should first tackle the existence of μ (mirroring Yang–Mills), possibly by casting IST axioms into a well-posed hyperbolic PDE or discrete dynamical system and proving an attractor. Next, establish coarse-graining results: adapt Glimm–DiPerna or kinetic theory techniques to the fractal setting. On the analytic side, one can investigate whether any known monokinetic limit theorems (Lions–Perthame, Golse, etc.) can be modified for deterministic Cantor distributions. Exploring the stochastic route, one could attempt a central-limit-type theorem for the fractal flow to justify effective diffusion. Finally, study the Wallstrom condition in models: see if and how the fractal topology might enforce quantized circulation (perhaps via homology of the attractor). A promising approach is a combined **variational/measure theory**

strategy: show the SRB measure *maximizes* a pressure functional that coincides with Guerra's action functional at the coarse-grained level.

In conclusion, we **cannot yet assert** that BM-IST is mathematically coherent in deriving the Bohmian velocity. It remains an intriguing hypothesis **with none of its key steps fully proven**. The outline above – bridging continuity equations, fractal calculus, and stochastic variational principles – is the most plausible path to a proof. Substantial new theorems are required, and until they are found, the question stays open.
